Predictive UFLS Ride-Through Extension via PMU Wavefront Analysis

A Gaussian Process Learning Framework for Inertia-Conditioned Protection Coordination

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Abstract

Conventional frequency ride-through protection for generating units is calibrated to fixed thresholds under NERC Reliability Standard PRC-024, without access to real-time information about how long a frequency excursion will last at a given location. When the excursion would resolve within the ride-through window, units may trip unnecessarily, deepening the disturbance rather than riding through it. This paper advances two contributions. First, it establishes that the Full Width at Half Maximum (FWHM) of the frequency wavefront, as observed at a local phasor measurement unit (PMU), encodes the predicted duration of the frequency excursion at that location — directly, in the same units, from the same PMU time-series. Second, it proposes a Gaussian process regression architecture that learns the local FWHM-to-duration mapping from accumulated event data, produces explicit uncertainty estimates, and activates ride-through extension only when the predicted excursion duration falls within the PRC-024 ride-through window under worst-case posterior uncertainty. PMU-to-PMU observation sharing via the existing IEEE C37.118 layer expands the calibration dataset without introducing real-time latency into the protection decision. The 2008 FPL Flagami event is reconstructed by simulation to assess consistency with the conjecture. A pathway for revision of NERC PRC-024 to accommodate model-based alternative compliance is proposed.

1. Introduction

The Flagami substation in Miami is not unfamiliar to one of this paper's authors. In the early 1980s, he worked at Florida Power & Light's Miami General office — close enough to Flagami to know the substation by name, though not by any direct engineering involvement. He moved on to AEP and then to BGE, and was at BGE when the February 2008 disturbance occurred. He came to NERC in 2009. It was there, around 2009 or 2010, while he and co-author Cummings were working on the NERC Aurora alerts, that the conversation took place. They were looking at the South Florida frequency records — the PMU data showing the disturbance propagating outward across the Eastern Interconnection, attenuating as it traveled through TVA and into Ontario, and reaching western Massachusetts [11]. The image that fixed itself was a bell: strike it at one point, and the vibration travels through the material, broadening and attenuating as it goes. The grid rang the same way.

The question they posed: if you could observe the shape of that ring at a local generating unit — whether the frequency excursion arriving at that location was sharp and brief or broad and already attenuating — could you tell the protection relay whether it actually needed to trip? If the excursion was broad at the local PMU, the unit might be sitting two seconds from riding through when the protection relay fired unnecessarily. That question has percolated for fifteen years. The analytical tools to answer it rigorously are now available.

The protection coordination problem at the heart of this paper is not exotic. It follows directly from what frequency protection relays do and do not know at the moment they make a trip decision.

A generating unit connected to the transmission system carries frequency protection settings consistent with NERC Reliability Standard PRC-024. Those settings define a frequency-versus-time envelope: if the local frequency falls below a specified threshold and remains there for a specified duration, the unit trips. The settings are fixed. They are conservatively calibrated — for the worst-case conditions the grid might present — because the relay has no way to know how long the local excursion will last when it makes the trip decision.

That conservatism has a cost. When the frequency excursion would resolve within the ride-through window — when the grid, left alone, would absorb the disturbance before the unit's PRC-024 time limit expires — units may trip anyway. Their trips further reduce generation, deepen the frequency excursion, trigger additional trips, and potentially convert a recoverable disturbance into a cascade. The relay's decision, made in ignorance of the excursion's duration, can make the grid less stable than it would have been had the relay known what the PMU was already recording.

This is the protection coordination gap. It is not a gap in relay hardware capability — modern digital relays can execute sophisticated algorithms in milliseconds. It is an information gap. The relay does not know how long the local frequency excursion will last when it needs to make the trip decision.

Two developments have changed the analytical landscape sufficiently to warrant revisiting this gap. The first is the widespread deployment of phasor measurement units across the transmission system. PMUs sample frequency, voltage, and current at rates of 30 to 60 frames per second, time-stamped to UTC via GPS. The temporal resolution is sufficient to observe the shape of a frequency wavefront as it arrives at a measurement point — not just the frequency value, but the wavefront's time-domain structure, including its width. The second is the maturation of machine learning methods for time-series regression that can operate on PMU data streams with explicit uncertainty quantification. The combination of high-resolution local measurement and a learning architecture that improves with experience opens a pathway to information that the relay previously lacked.

This paper proposes that the Full Width at Half Maximum (FWHM) of the frequency wavefront — a shape characteristic computable from the existing PMU time-series without additional instrumentation — predicts the duration of the local frequency excursion below the PRC-024 ride-through threshold. FWHM and excursion duration are both measured in seconds from the same PMU recording; the GP model learns the local mapping between them from accumulated event data, without requiring a closed-form derivation for the specific network topology.

The Flagami event in February 2008 serves as the motivating case study. A 345 kV bus fault at the Flagami substation in Miami initiated a cascade of generating unit trips, including Turkey

Point nuclear units 3 and 4, through a sequence of frequency protection operations. Co-author Cummings documented the wave-like propagation of the resulting frequency disturbance across the Eastern Interconnection using PMU data — the first such observation on an interconnection-wide basis [11]. Post-event analysis indicated that inertia conditions at some of the tripped units were consistent with successful ride-through, had protection settings permitted it. The simulation reconstructs the Florida network and assesses whether the FWHM of the wavefront at the tripped-unit locations is consistent with the predicted excursion duration.

A third development reinforces the paper's timeliness. The rapid penetration of inverter-based resources — utility-scale wind, solar photovoltaic, and battery storage operating with grid-following inverter controls — is reducing and destabilizing aggregate synchronous inertia across all North American interconnections. Simultaneously, the growth of hyperscale computational loads — data center campuses drawing hundreds of megawatts from the transmission system — is introducing a new class of large, abrupt generation-load imbalance events that the frequency protection framework was not designed to accommodate. NERC's 2026 alert on large computational load disconnections formally identified this combination as a grid reliability concern. The consequence is an environment where effective inertia is lower, more variable by hour and season, and more consequential for individual frequency events than it was when PRC-024 was calibrated. The case for duration-conditioned ride-through extension strengthens as that mismatch grows.

The standards implication is a proposed revision pathway for NERC PRC-024 that accommodates a model-based alternative compliance path alongside the existing fixed-envelope approach. The near-term deliverable is a NERC Standards Authorization Request (SAR). Section 7 develops the standards pathway and identifies the appropriate standards development vehicles.

2. Background

2.1 Frequency Wavefront Propagation

A generation loss event does not produce an instantaneous, spatially uniform frequency response across an interconnected grid. The frequency perturbation propagates as an electromechanical wave, traveling outward from the disturbance origin through the network of synchronous machines and transmission links. The propagation speed and attenuation characteristics depend on the distribution of rotating machine inertia, network impedance, and load characteristics throughout the grid. At locations near the disturbance's origin, the wavefront arrives quickly, and the frequency deviation is sharp. At more distant locations, the wavefront arrives later and, if the intervening inertia is sufficient, with a reduced rate of change and a broader time-domain shape.

This spatial and temporal structure of the frequency response has been documented in the PMU literature since the early deployment of wide-area measurement systems. The Western Interconnection, with its large geographic extent and heterogeneous inertia distribution, provides the most extensively studied examples. Electromechanical wave propagation speeds on the order of hundreds to thousands of kilometers per second have been estimated from PMU event recordings, with attenuation characteristics that reflect the aggregate inertia along the propagation corridor.

2.2 Rate of Change of Frequency as an Inertia Estimator

The conventional approach to inertia estimation in protection and monitoring applications uses ROCOF — the Rate of Change of Frequency, df/dt — measured at the local PMU or derived from the relay's frequency tracking. The relationship between ROCOF and inertia follows from the swing equation. For a machine observing a generation-load imbalance ΔP, the initial ROCOF is inversely proportional to the effective system inertia constant H. This relationship is analytically grounded and has been the basis for inertia estimation in grid studies.

In practice, ROCOF estimation in the first seconds of a disturbance is challenged by three factors. First, electromechanical oscillations superimposed on the frequency trajectory introduce measurement noise, rendering instantaneous ROCOF estimates unreliable without filtering. Second, the filtering required to stabilize ROCOF estimates introduces delay — typically 100 to 500 milliseconds — during which the protection decision may need to be made. Third, ROCOF is a scalar derived from the local frequency time series and contains no spatial information about how the disturbance is propagating through the network. As a result, ROCOF-based inertia estimation is least reliable precisely when it is most needed: in the first fraction of a second of a severe disturbance.

2.3 FWHM in Signal Processing Context

Full Width at Half Maximum is a standard measure of pulse width in signal processing, spectroscopy, and optics. For a pulse or peak, FWHM is the duration of the pulse at half its maximum amplitude. It is a shape characteristic, not an amplitude characteristic — two pulses with identical maximum amplitude but different widths have different FWHM values.

Applied to a frequency wavefront, FWHM is the duration of the frequency deviation event — measured from the local PMU frequency time-series — at half the maximum frequency deviation from nominal. Two disturbances with the same maximum frequency deviation but different propagation histories will produce different FWHM values at a given observation point. A disturbance traveling through a high-inertia corridor produces a broader, slower wavefront — larger FWHM — than the same disturbance traveling through a low-inertia corridor. FWHM is directly computable from the existing PMU frequency time-series, requires no additional instrumentation, and does not require knowledge of the disturbance origin location or the network topology between origin and observer.

2.4 UFLS and the Cascade Dynamic

Under-Frequency Load Shedding (UFLS) is the last-resort defense against frequency collapse. When frequency falls below programmed thresholds, UFLS relays automatically disconnect blocks of load, reducing demand to arrest the decline in frequency. UFLS is designed as a backstop — it operates after the primary frequency response from spinning reserve and governor action has had time to take effect. It sheds load in blocks, in sequence, each block triggered at a successively lower frequency threshold.

The interaction between the generating unit frequency protection trips and the UFLS defines the cascade dynamics. When generating units trip on frequency protection, they reduce supply further — deepening the frequency excursion and activating additional UFLS blocks. If generating unit trips are unnecessary — if the unit would have remained within its ride-through envelope had inertia conditions been correctly assessed — those trips consume UFLS headroom and widen the outage without providing protective benefit. Preventing unnecessary protection trips in high-inertia conditions directly reduces UFLS demand and improves the probability of frequency recovery.

2.5 Inverter-Based Resources, Large Computational Loads, and the Inertia Variability Problem

The frequency protection coordination problem described in this paper is not static. Two structural trends in the bulk electric system are simultaneously making it more severe and more consequential.

The first is the large-scale displacement of synchronous generation by inverter-based resources (IBRs). Utility-scale wind and solar photovoltaic plants, and battery storage systems operating under grid-following inverter controls, do not contribute synchronous inertia to the interconnection. A grid-following IBR observes the grid frequency and modulates its power output accordingly; it does not exchange kinetic energy with the grid through the electromagnetic coupling of a rotating mass. As IBR penetration increases and synchronous generators are decommissioned or operated less frequently, aggregate system inertia decreases and — crucially — becomes more variable. Inertia is now a function of dispatch conditions: high during periods of synchronous-machine-dominated dispatch, low during periods of high renewable penetration. A given generating unit at a given network location may face dramatically different effective inertia conditions at different hours of the day or different seasons of the year. Fixed PRC-024 settings, calibrated to a single conservative envelope, cannot track this variability.

Grid-forming inverter technology offers a partial response to this problem. A grid-forming IBR synthesizes an inertia-like response — it emulates the kinetic energy exchange of a synchronous machine through fast power electronics control — and can contribute to frequency support in ways that grid-following IBRs cannot. Grid-forming deployments are growing but remain a small fraction of total IBR capacity. Their contribution to inertia is synthetic rather than mechanical, and its interaction with the electromechanical wave propagation that produces the FWHM signal is a refinement addressed in Section 3.2.

The second trend is the emergence of hyperscale computational loads as a transmission-connected load class. Data center campuses drawing 100 to 500 megawatts from the bulk electric system are now interconnecting at transmission voltage in multiple North American markets, including ERCOT. These facilities can connect, disconnect, or shift load in large blocks over timescales of seconds to minutes — timescales directly relevant to frequency stability. A facility trip or a large workload migration event is, from the grid's perspective, a sudden generation-load imbalance of the same character as a large generating unit trip. The frequency protection system sees it as a disturbance and responds accordingly.

NERC's 2026 alert on large computational load disconnections formally identified this load class as a grid reliability concern. The alert documents the frequency of consequences from large computational-load disconnection events and their interactions with existing frequency protection systems, including instances in which UFLS was activated. This is the regulatory anchoring event for the standards pathway developed in Section 7: NERC has now formally identified the problem class; the question before the standards community is what protection coordination framework is appropriate in response.

The combination of these two trends — lower and more variable synchronous inertia from IBR penetration, and more frequent large-block frequency perturbations from hyperscale load dynamics — creates a threat environment for which fixed PRC-024 settings are increasingly poorly calibrated. The misalignment between static protection settings and a dynamically variable inertia environment is the structural gap that this paper's proposed provisions pathway addresses. It also, incidentally, creates a more data-rich environment for the GP calibration architecture: more frequent disturbance events mean more FWHM observations, faster model convergence, and a training dataset that reflects the current inertia regime rather than a historical average.

3. Physical Foundation and Grid Application

3.1 Established Physical Basis

The principle that the Full Width at Half Maximum of a propagating wavefront encodes properties of the medium through which it has traveled is established across multiple branches of physics, and in each case rests on the same underlying mechanism: dispersion or attenuation accumulates along the propagation path. The temporal or spectral width of the observed feature at the downstream measurement point reflects that accumulated effect.

In spectroscopy, thermal Doppler broadening produces a Gaussian absorption line profile whose FWHM is proportional to the square root of the radiating medium's temperature. Pressure broadening produces a Lorentzian profile whose FWHM encodes the collision frequency of the absorbing species and hence the local gas density. The convolution of both mechanisms — the Voigt profile — carries information about both temperature and pressure simultaneously [1]. In quantitative spectroscopy, the FWHM of the observed line profile is not merely a descriptor of line shape; it is the primary observable from which the medium's physical state is extracted.

In a dispersive single-mode optical fiber, a short pulse broadens temporally as it propagates, and the temporal FWHM of the output pulse encodes the fiber's group velocity dispersion parameter integrated over the path length [2]. In ultrasonics, pulse FWHM broadening in viscoelastic media encodes frequency-dependent attenuation coefficients — and hence the medium's viscosity and elastic moduli — and underlies tissue characterization in diagnostic ultrasound and material characterization in industrial non-destructive evaluation [3]. In each domain, FWHM is a path-integrated quantity: it accumulates information about the medium along the entire propagation corridor, not solely at the observation point.

The formal analogy between these established results and frequency wavefront propagation in large power systems was drawn by Thorp, Seyler, and Phadke [4], who demonstrated that frequency disturbances propagate through the bulk electric system as electromechanical waves. Using a continuum approximation of the network, they showed that propagation characteristics are governed by the spatial distribution of machine inertia and network impedance — directly analogous to the group-velocity dispersion in a fiber or the viscoelastic attenuation in an acoustic medium. The inertia distribution of the grid serves as the medium's dispersive character; a downstream PMU's frequency observation carries information about the inertia environment integrated over the propagation corridor from the disturbance's origin. Co-author Cummings documented the wave-like propagation of the February 2008 South Florida frequency disturbance across the entire Eastern Interconnection using PMU data — the first interconnection-wide empirical observation of this phenomenon [11] — providing direct evidentiary grounding for the Thorp et al. framework in a real-world event.

The contribution of this paper is the application of this established physical principle to the protection coordination problem, with a specific and direct connection to PRC-024's ride-through requirement. FWHM is itself a duration — measured in seconds from the PMU frequency time-series — expressing the width of the frequency deviation event at half its maximum amplitude. The PRC-024 ride-through requirement is also expressed in duration: the time a unit must remain connected at a frequency below a specified level. The GP model therefore learns a duration-to-duration mapping, from the FWHM of the incoming wavefront to the predicted duration of the excursion below the ride-through threshold. Both quantities are measured in seconds, from the same PMU recording, without intermediate estimation steps. For a frequency excursion with approximately Gaussian temporal profile — the shape that propagation through an inertia-distributed medium tends to produce [4] — FWHM and threshold-exceedance duration are monotonically related by a transformation that depends only on the ratio of the threshold level to the maximum frequency deviation. The GP learns this relationship locally, without requiring the transformation to be derived analytically for a specific network topology. The domain-specific claim is that this mapping is sufficiently stable at a given PMU location under typical operating conditions that a locally calibrated GP model can reliably exploit it for ride-through decisions.

3.2 Physical Motivation

The physical basis for the FWHM-to-duration encoding follows directly from Thorp et al.'s electromechanical wave framework [4]. A synchronous machine with high inertia resists acceleration and deceleration — it exchanges kinetic energy with the grid slowly and acts as a damping element on the propagating wavefront. A propagation corridor containing machines with high aggregate inertia therefore slows and broadens the wavefront, producing a longer-duration frequency deviation at downstream observation points compared to what the same disturbance would produce propagating through a low-inertia corridor. Critically, a broader wavefront means the frequency excursion at the observation point is not only wider in FWHM — it is also attenuating faster and recovering sooner. The unit experiencing a broad, attenuating wavefront is in a fundamentally different situation from one experiencing a sharp, sustained one: the former is likely to ride through; the latter may not. FWHM captures this distinction directly, in the same temporal units as the PRC-024 ride-through requirement.

The FWHM of the observed wavefront integrates this inertia effect over the entire propagation corridor. It is not a point measure of local conditions — it is a path-integrated proxy for how the medium has dispersed the disturbance between the origin and the PMU. This property makes it analytically distinct from ROCOF, which estimates inertia from the local machine's instantaneous swing dynamics and contains no information about the propagation corridor or the excursion's likely duration.

A secondary physical effect reinforces the expected relationship: load damping. High-load corridors, which tend to coincide with high-inertia transmission regions, also exhibit stronger load-frequency damping. Both mechanisms — machine inertia and load damping — broaden the wavefront and shorten the excursion duration in the same direction. The FWHM signal integrates both effects.

The growth of IBR penetration introduces a scoping condition that must be stated explicitly. Grid-following IBRs do not contribute to electromechanical wave propagation — they do not exchange kinetic energy through rotating mass coupling and therefore do not participate in the wavefront dynamics that produce the FWHM signal. FWHM, as used in this paper, denotes the synchronous inertia within the propagation corridor. In a high-IBR environment, synchronous inertia may be a conservative lower bound on total effective inertia — grid-forming IBRs contribute synthetic inertia that partially supplements the synchronous component — but the FWHM signal does not capture that synthetic contribution directly. For a protection application, this conservatism is in the correct direction: ride-through extension activates only when synchronous inertia alone is sufficient, without crediting synthetic inertia whose availability depends on IBR dispatch. We note that whether grid-forming IBR synthetic inertia leaves a detectable signature in the wavefront shape that a GP model could learn to identify is a follow-on research question and is deferred.

3.3 Why Closed-Form Derivation Is Not the Goal

Deriving an analytical FWHM-to-inertia relationship for a general network topology requires solving the frequency wave equation on an impedance-distributed graph with spatially varying inertia and damping coefficients. This is not tractable in closed form for real network topologies: the wave equation is linear in a homogeneous medium, but the grid is neither spatially homogeneous nor topologically regular. Network impedances, machine characteristics, and load distribution introduce spatial heterogeneity, rendering the general solution analytically intractable.

The conjecture does not require solving this problem. It requires only that the encoding relationship exist and be learnable from local observations. This is the appropriate scope for the claim. The closed-form relationship is a path toward a universal model that generalizes across all network topologies. The conjecture and the learning architecture described in Section 4 are a path toward a locally calibrated model — one that knows this network, in this configuration, under typical operating conditions, and improves its estimates as it observes more local events. The locally calibrated model is more directly deployable and does not require solving the general problem.

3.4 The Local Encoding Hypothesis

The deployability hypothesis underlying the conjecture is that the FWHM-to-inertia mapping, while not analytically derivable for general topologies, is sufficiently stable at a given network location — over a typical range of operating conditions — that a model observing accumulated local event data can learn a useful approximation. This hypothesis has three implications for the learning architecture.

First, the model is location-specific. A model trained on events observed at one PMU does not generalize without recalibration to a PMU at a different network location, because the inertia distribution relevant to a ride-through decision at any location is the inertia in the corridor between typical disturbance origins and that specific location.

Second, the model improves with experience. Every disturbance the PMU observes — with known outcome and known frequency trajectory — is a training sample that refines the model's estimate of the local FWHM-to-inertia mapping. The model drifts toward local calibration as the event record accumulates.

Third, the model has a sensible prior. Before the model has observed any local events, the FWHM-to-inertia relationship can be approximated from a simplified network model using available network parameters. This prior is not accurate — it ignores topology-specific effects —

but it is physically motivated and, by design, conservative, providing a reasonable and safe starting point.

4. A Learning Architecture for Local Duration Estimation

4.1 Gaussian Process Regression

The learning architecture proposed for the local excursion duration estimation model is Gaussian process (GP) regression. A Gaussian process defines a distribution over functions — in this application, a distribution over possible FWHM-to-duration mappings. Given training data consisting of observed FWHM values and the associated measured excursion durations below the PRC-024 threshold, the GP updates this distribution to produce a posterior that concentrates on mappings consistent with the observed data.

GP regression was selected for three properties that are essential for deployment in a protection system context. First, explicit uncertainty quantification: at any input FWHM value, the GP posterior provides a full posterior distribution — mean and variance — from which the probability that excursion duration will fall within the ride-through window can be directly computed. This property is essential for the conservative activation logic described in Section 4.4. Second, interpretability: the GP model state at any time is fully inspectable; the kernel function and its learned hyperparameters are visible and auditable. Third, bounded behavior: the GP posterior is constrained by the prior and cannot drift arbitrarily from physically motivated bounds regardless of what the training data contains.

4.2 Kernel Selection and Physics-Based Prior

The GP kernel encodes the prior expectation about how similar FWHM values should produce similar predicted durations. A squared exponential kernel with a learned length scale is the natural choice: it asserts that nearby FWHM values produce similar duration estimates, with the length scale governing how rapidly the similarity decays. The kernel hyperparameters — amplitude and length scale — are initialized from the physics-based prior and updated as training data accumulates.

The physics-based prior encodes the expected FWHM-to-duration relationship before any local observations are made. It is constructed from the monotonic relationship described in Section 3.1: wider FWHM implies shorter excursion duration below threshold. The prior asserts this monotonically decreasing relationship, parameterized conservatively: for any observed FWHM, the prior mean assigns a duration estimate at the upper end of the physically plausible range for the local network class. This conservatism is intentional — a conservative prior means that before the model has accumulated sufficient local observations, the posterior duration estimate will exceed T

This improvement property is bounded. The GP hyperparameters are constrained to remain within physically plausible ranges throughout the training process. Kernel amplitude is bounded by the plausible range of excursion durations for the network class; length scale is bounded by the range of FWHM values consistent with physically realistic wavefronts. The model cannot drift arbitrarily from the physics prior regardless of what the training data contains.

4.3 Training Data and Drift Toward Improvement

Each disturbance the PMU observes provides a training sample. The sample consists of: (1) the observed FWHM of the frequency wavefront, computed directly from the PMU frequency time-series; and (2) the measured duration of the frequency excursion below the applicable PRC-024 threshold at the local PMU, also read directly from the same PMU recording. No ROCOF estimation, no post-event filtering, and no secondary calculation is required. The training label is a directly observable quantity in the same units as the input — both are durations in seconds from the same time-series. This simplicity is a significant architectural advantage: the ground-truth label no longer requires ROCOF-based estimation, which introduced filtering delay and measurement noise and was the weakest link in an inertia-mediated formulation.

As training samples accumulate, the GP posterior narrows. The model's uncertainty about the FWHM-to-duration mapping at the local PMU location decreases. The ride-through extension activates more frequently in high-inertia conditions because the posterior duration estimate falls within the ride-through window with greater confidence. The model drifts toward local calibration: it learns the relationship between wavefront width and excursion duration as it manifests at this specific network location, under the operating conditions typical for this facility.

4.4 Conservative Activation Logic

The following notation is used in this section: $FWHM_{obs}$ denotes the observed FWHM at the local PMU; μ_T and σ_T are the GP posterior mean and standard deviation of the predicted excursion duration at that FWHM value; k is a conservatism parameter set by the utility; and $T_{limit}(f)$ is the PRC-024 ride-through time limit at the observed frequency level f — read directly from the standard's frequency-time curve, not a learned parameter.

Extension activates when $\mu_T(FWHM_{obs}) + k \cdot \sigma_T(FWHM_{obs}) < T_{limit}(f)$. In plain terms: the model's predicted excursion duration, inflated by k standard deviations of its own uncertainty, must still fall within the PRC-024 ride-through window. Wide uncertainty inflates the upper bound, making it harder to remain below $T_{limit}(f)$, so extension does not activate. The architecture remains self-conservatizing: ignorance expresses itself as inaction.

Note that the conservatism threshold operates in the opposite algebraic direction from an inertia-based formulation. Estimating inertia requires subtracting uncertainty to ensure the lower bound exceeds a minimum threshold. Estimating excursion duration requires adding uncertainty to ensure the upper bound remains below a maximum limit. The underlying logic is identical —

demonstrate safety under worst-case uncertainty — but the direction follows the output variable. The duration formulation has a further advantage: $T_{limit}(f)$ is not derived or estimated. It is read from the PRC-024 frequency-time curve at the current measured frequency, making the threshold unambiguous and fully auditable.

When *k* is large, the condition is difficult to satisfy — extension activates only when the model is highly confident that the excursion will resolve within the ride-through window. When *k* is small, the threshold is more readily met. The default value of *k* should be set conservatively during initial deployment and relaxed only as the model accumulates a validated event record. In all cases where the condition is not satisfied — including all cases where model uncertainty is wide — the existing PRC-024 protection setting governs.

4.5 Verification, Validation, and Auditing Requirements

Continuous model updating introduces a verification and validation challenge that must be addressed in the standards pathway. The proposed V&V framework has four elements.

Parameter bounds enforcement. GP hyperparameters are bounded within ranges derived from physically plausible network models for the local facility class. Updates that would push parameters outside these bounds are rejected. The bounds are specified at deployment and are not subject to field modification without a formal protection settings change process.

Holdout validation. A fraction of observed events — nominally twenty percent — is withheld from training and used to validate the model's duration predictions against directly measured excursion durations from the withheld events. The model's root mean square error on the holdout set is logged continuously. If holdout RMSE exceeds a utility-specified threshold, ride-through extension activation is suspended automatically pending manual review by the protection engineering staff.

Auditability. The model state — current kernel parameters, training data summary statistics, posterior mean function, holdout validation results, and the activation log — is continuously logged and available for inspection. The protection engineer can see exactly what the model has learned and review every instance in which extension was or was not activated.

Update rate limiting. Model parameter updates are performed on a scheduled basis — daily or weekly, depending on disturbance frequency at the location — rather than in real time following each event. Each update batch is subject to the holdout validation check before parameters are committed. Real-time update is explicitly excluded from the architecture.

4.6 Local Sovereignty: A Protection Function That Works When Everything Else Has Failed

The architecture described in this section satisfies a design requirement that is distinct from, and more fundamental than, the V&V requirements in Section 4.5. A UFLS event is by definition a grid stress condition in which normal operational dependencies have been compromised.

External power may be degraded. Wide-area communications may be congested or unavailable. Control room operators may be managing multiple simultaneous events. Any protection function whose decision depends on external power, external communications, or operator action is precisely the kind of function that may fail under UFLS conditions.

The adjustable UFLS PMU architecture described in this paper is designed to satisfy what this document terms the local sovereignty principle: a protective system must be capable of making and executing its ride-through decision without requiring external power, external communications, or operator action in the decision loop. Every element of the architecture that touches the real-time protection decision is locally resident and locally powered. The FWHM computation runs on the local PMU's processor. The GP posterior is resident in local memory. The activation threshold comparison is a local arithmetic operation. The ride-through extension, if activated, is a local relay action. Nothing in the decision path requires a signal from outside the device.

This is not a novel design principle in protective relay engineering. The Aurora protection device — developed to protect rotating machinery against the out-of-synchronism damage mode demonstrated in the 2007 Idaho National Laboratory test — was designed on the same principle. The attack scenario is precise: the adversary opens a breaker and intends to reclose it onto an out-of-synchronism grid in approximately 15 to 20 cycles, inducing the rotational stress that can destroy a large generator. The Aurora device reads the local grid physics and, when it detects an out-of-synchronism condition, lifts the close coil lead — making reclose mechanically impossible without any trip command, operator action, or communication with a remote system. The protection lies in preventing the injurious state transition, not in responding after it has occurred. The device works because it is purely local: it observes what the local grid is doing and acts on that observation alone.

The UFLS ride-through extension architecture operates on the identical design principle, in mirror-image logic. Where the Aurora device prevents an injurious close, the UFLS device prevents an unnecessary open. Both read local physics — the Aurora device reads grid synchronism; the UFLS device reads wavefront shape — and both act locally to prevent a bad state transition from occurring. The locally sovereign protection function is the answer to threat conditions, whether adversarial or operational, in which the normal dependency chain cannot be relied upon.

UFLS conditions are the frequency stability analog. When frequency falls below UFLS thresholds, the grid is already under severe stress. The protection decision must be made in the first hundreds of milliseconds. Communications congestion, operator workload, and potential power supply degradation at the substation level are all elevated. A ride-through extension architecture that depends on any of these factors in the decision loop is not fit for purpose in the conditions it is designed to address. The local sovereignty design principle is not optional for this application — it is the enabling condition that makes the architecture credible.

The implications for standards of the local sovereignty framing are developed in Section 7.3. For the architecture, the consequence is a clear and non-negotiable constraint: the GP model, the activation logic, and the relay action must all be executable from local power and local state, with no real-time external dependency. The C37.118 calibration loop, the holdout validation, and the audit logging are all post-event processes that may use external communications — but none of them are in the decision path.

5. PMU-PMU Observation Sharing as Calibration Data Expansion

5.1 The Calibration Data Constraint

A significant practical constraint on the GP model is the rarity of large disturbances at any given network location. The model's calibration quality depends on the number of training samples — FWHM observations with associated inertia estimates — accumulated at the local PMU. Events sufficient to produce a measurable FWHM signature may occur at a given location only a few times per year. At that rate, local-only model calibration converges slowly, and the period during which the model operates near its conservative prior — with wide uncertainty and infrequent activation of extensions — is prolonged.

5.2 C37.118 as Training Data Transport

IEEE C37.118 is the standard protocol for synchrophasor data communication in wide-area measurement systems. Every PMU in a utility's measurement network transmits frequency, voltage, and current phasor data at 30 to 60 frames per second to a phasor data concentrator (PDC) via C37.118. This infrastructure is already deployed across the transmission system for monitoring and post-event analysis. No new communication infrastructure is required for the PMU-PMU sharing concept.

The proposal is to use this existing infrastructure as a training data transport for the GP calibration loop. When a disturbance occurs, every PMU in the measurement network that observes a measurable frequency event records a FWHM value at a different point on the propagating wavefront. Those FWHM observations, paired with post-event inertia estimates derived from the full frequency trajectory, are training samples. Shared via C37.118 after the event, they expand the local model's training dataset with observations from disturbances that may have produced a stronger FWHM signature at other network locations than at the local PMU.

5.3 The Calibration Loop Is Not the Decision Loop

A critical architectural distinction separates the real-time protection decision from the calibration loop. The ride-through extension decision is made locally, in milliseconds, from the FWHM of the locally observed frequency wavefront and the current GP posterior. No C37.118 traffic is in the decision loop. No neighbor PMU data is required in real time. The latency characteristics of C37.118 transmission over utility-wide-area networks — typically 100 milliseconds or more — are irrelevant to the protection decision.

Neighbor PMU data arrives in the calibration loop after the event. Post-event FWHM observations from neighboring PMUs are transmitted to the local PDC, aggregated, and processed on the training data update schedule. They expand the local model's training dataset without introducing any dependency on communication latency in the decision path. The local decision remains fully local.

5.4 What Neighbor Data Adds to Model Quality

The value of neighbor PMU observations lies in the spatial coverage of the inertia landscape surrounding the local PMU. A disturbance originating near a distant neighbor PMU may produce a FWHM signal too small to be useful at the local PMU — the disturbance has attenuated over the long propagation path — while producing a clear signal at PMUs close to the origin. Those close-in PMU observations, shared via C37.118, inform the local model about how disturbances from that direction propagate through the intervening network.

Over many events from many directions, the local model accumulates a spatial picture of the effective inertia landscape. The improvement is not uniform: disturbances from directions well-represented in the training data calibrate the model more precisely than disturbances from novel or rare directions. The uncertainty quantification in the GP posterior reflects this: well-represented FWHM regimes have narrow posterior uncertainty, whereas poorly represented regimes retain wider uncertainty and are less likely to trigger the conservative activation threshold. The architecture self-corrects toward conservatism in situations where the model has less evidence.

6. The Flagami Case Study

6.1 Event Summary

On February 26, 2008, a fault at the FPL Flagami 345 kV substation in Miami, Florida initiated a cascade of generating unit trips across the southern Florida peninsula. The fault caused a voltage excursion on the 345 kV transmission system serving the Miami metropolitan area. Turkey Point nuclear generating station — located approximately 40 miles south of the Flagami substation — experienced a loss of off-site power supply, initiating automatic reactor trips on both units 3 and 4. Multiple additional fossil generating units tripped on frequency and voltage protection as the cascade propagated through the Florida network. Total generation loss exceeded 4,000 MW at the cascade peak. Approximately 1.5 million customers in the FPL service territory experienced outages lasting up to several hours.

The event is documented in NERC's post-event disturbance analysis and the FERC/NERC joint inquiry report, which examined the interactions among protection systems that contributed to the cascade's severity. Post-event analysis indicated that inertia conditions in the affected network region at the time of the event were consistent with successful ride-through for some of the generating units that tripped on frequency protection — meaning that, under inertia conditions sufficient for frequency recovery, those units need not have tripped to protect themselves.

This finding is the evidentiary foundation for the Flagami case study as applied to the conjecture. The simulation does not attempt to demonstrate that the conjecture's algorithm would have prevented the cascade. It asks a more limited question: is the FWHM signature of the simulated frequency wavefront at the tripped unit locations, at the moment the protection decision would have been made, consistent with what a calibrated GP model would have recognized as a high-inertia condition?

6.2 Simulation Methodology

The Flagami network reconstruction uses the Florida sub-network derived from NERC's power flow and dynamics model database, calibrated to 2008 system conditions using load flow data and generator dispatch records available from NERC's post-event documentation. The simulation is conducted on [platform to be confirmed by Dagle/PNNL]. The fault is initiated at the Flagami 345 kV bus, consistent with the documented fault event. The simulation captures the resulting voltage excursion, the frequency transient propagation across the Florida network, and the cascading unit trip sequence.

Frequency time-series are extracted from the simulation at each generating unit location at PMU-equivalent sampling rates of 60 frames per second. FWHM is computed from the extracted frequency time-series at each location using the following procedure: the frequency deviation from the 60 Hz nominal is computed; the maximum deviation is identified; the half-maximum level is established; and the duration of the frequency event at the half-maximum level — the interval between the two time points at which the deviation crosses the half-maximum level — is measured. This FWHM value is the input to the illustrative GP model inference.

The simulation is stated as a reconstruction from available network data, not as a validated model. The network model is an approximation of 2008 conditions; FWHM is computed from simulated, not measured, frequency time series; and the GP model is illustrative rather than trained on actual Florida grid-event data. The simulation is designed to test consistency with the conjecture, not to validate the GP model's deployment-readiness.

6.3 Consistency Assessment Framework

The consistency assessment asks whether the simulated FWHM values at the generating unit locations fall within the range predicted by the conjecture for high-inertia conditions at the time of the event. Specifically, the assessment compares the simulated FWHM values at units that post-event analysis identified as having inertia conditions consistent with ride-through with those at units that clearly required tripping. If the conjecture is correct, the former should produce systematically larger FWHM values than the latter.

The GP model used for the consistency assessment is initialized from the physics-based prior with no local training data, representing the model's state at initial deployment. The posterior inertia estimates at the observed FWHM values are compared against the conservative activation threshold. The consistency question is: does the posterior estimate, even under wide prior uncertainty, indicate a high-inertia signal at the units identified as ride-through-capable in the post-event analysis?

6.4 Simulation Results

We will complete the quantitative results — FWHM values at each unit location, corresponding GP posterior estimates, comparison with the conservative activation threshold, and the implied activation decision — once we receive the PNNL simulation results. This section is reserved for that material.

7. Standards Pathway

7.1 PRC-024 Current Structure

NERC Reliability Standard PRC-024-3, effective 2020, establishes frequency ride-through requirements for generating facilities interconnected to the bulk electric system. The standard requires that generating facilities remain connected when frequency is within the bounds established by the standard's frequency-time curves — the envelope within which units must not trip due to frequency protective relay action. PRC-024 applies to generating facilities interconnected at or above 100 kV that have programmable protection systems. It was developed based on the historical frequency excursion profile of the North American interconnections and sets separate curves for underfrequency and overfrequency conditions.

The standard's compliance architecture is fixed-threshold: a facility demonstrates compliance by demonstrating that its protection settings do not trip within the ride-through envelope. There is no provision for a site-specific, condition-adaptive, or model-based approach. The envelope is conservatively calibrated — for the worst-case inertia conditions the interconnection might present — and is applied uniformly regardless of the actual inertia conditions at the time of any specific event.

7.2 The Gap in the Provisions Pathway Addresses

Fixed-threshold compliance does not distinguish between low-inertia and high-inertia conditions. The PRC-024 envelope was calibrated for conditions where generating unit trips would worsen the frequency excursion — conditions of low aggregate inertia, where each additional trip deepens the frequency decline. When aggregate inertia is high, the frequency excursion is more likely to be self-correcting, and trips of generating units within the PRC-024 envelope represent a lost ride-through opportunity without providing a protective benefit.

The result is a systematic conservatism bias that is most costly precisely when inertia is highest: under conditions where the grid is most resilient, protection settings engineered for worst-case inertia still trip units unnecessarily. As renewable generation displaces synchronous machines, aggregate inertia becomes more variable by hour, season, and dispatch condition. The variance in inertia conditions is increasing; the fixed PRC-024 envelope was calibrated for a lower-variance environment.

NERC's 2026 alert on large computational load disconnections provides current institutional grounding for the gap argument. The alert identifies large hyperscale load events as a reliability concern, specifically because of their interaction with frequency protection systems in a low-inertia environment — the same interaction that this paper's provisions address. The alert represents NERC's formal acknowledgment that the existing frequency protection framework requires examination in light of the changing inertia environment. A Standards Authorization Request filed contemporaneously with or shortly after the alert has a stronger institutional rationale than one filed without that evidentiary anchor. The alert should be explicitly cited in the SAR as the reliability concern that motivates the PRC-024 revision scope.

7.3 Proposed Provisions Pathway Structure

The provisions pathway within PRC-024 would allow a generating facility to demonstrate compliance through a model-based approach—specifically, a calibrated GP model that meets specified V&V requirements—as an alternative to fixed threshold settings. Under this approach, the facility's protection system is never less protective than the existing PRC-024 envelope: the model-based extension is strictly additive. In high-inertia conditions where the GP posterior satisfies the conservative activation threshold, the facility remains connected longer, improving system reliability. In all other conditions, the existing envelope governs.

The local sovereignty architecture described in Section 4.6 is a prerequisite for the provisions pathway, not merely a desirable feature. A model-based compliance demonstration that depends on external communications or operator action in the real-time decision loop cannot be relied upon under grid-stress conditions in which PRC-024 ride-through requirements apply. The provisions pathway must therefore include a local sovereignty requirement as a mandatory architectural condition: the ride-through extension decision and its execution must be achievable from local power and the local state alone. This requirement distinguishes the proposed architecture from supervisory override schemes and settings management systems, both of which introduce external dependencies in the decision path.

Four conditions are required to claim model-based compliance under the provisions pathway:

Model qualification. The GP model meets specified accuracy requirements — holding out RMSE below a threshold calibrated to the facility's consequence severity — as documented in an annual validation report submitted to the regional reliability coordinator.

Conservative activation threshold. The model's activation logic is set such that extension never activates unless the posterior inertia estimate, net of the conservatism factor k, exceeds the effective inertia level required for frequency recovery at the facility's network location.

V&V requirements. The facility maintains the parameter bounds enforcement, holdout validation, auditability, and update rate limiting described in Section 4.5.

Non-degradation. The model-based approach produces protection behavior no less conservative than the existing PRC-024 fixed envelope in expectation across the historical disturbance record for the facility's network location, as demonstrated by a retrospective analysis over the training event record.

7.4 Standards Development Pathway

The near-term deliverable is a NERC Standards Authorization Request (SAR) proposing revision of PRC-024-3 to include a provisions pathway for inertia-conditioned ride-through extension. The SAR would be filed with NERC's Standards Department and would initiate the standards development process under NERC's reliability standards development procedure. This paper and the PNNL simulation results constitute the technical evidentiary foundation for the SAR.

The appropriate drafting team for PRC-024 revisions is the Protection System Reliability Standards Drafting Team. The SAR would request that the provisions pathway be developed as an alternative compliance path within PRC-024-4 or as a companion standard. Bob Cummings is the identified contact for the NERC standards engagement.

The IEEE Power System Relaying and Control Committee (PSRC) is the appropriate standards body for relay-level implementation requirements — specifically, the V&V framework, parameter bound enforcement, and audit logging described in Section 4.5. Coordination between the NERC standards process and IEEE PSRC is advisable to ensure that the technical requirements at the relay level are consistent with the reliability requirements at the standards level. A joint technical paper submitted to the IEEE PSRC, concurrent with the NERC SAR, would initiate that coordination.

Conclusion

This paper advances two contributions to the frequency protection coordination problem, rooted in a conversation near the Flagami substation about the grid ringing like a bell.

The first contribution is the application of an established physical principle — that FWHM of a propagating wavefront encodes properties of the medium through which it has traveled — to the specific and direct problem of predicting excursion duration at a local PMU. FWHM is itself a duration. The PRC-024 ride-through requirement is expressed in duration. The GP model learns a duration-to-duration mapping from the width of the incoming wavefront to the predicted time at which the local frequency will remain below the ride-through threshold. Both quantities are measured in seconds, from the same PMU recording, without intermediate estimation. The Thorp et al. framework establishes the physical basis [4]; co-author Cummings's PMU documentation of the South Florida disturbance provides the empirical grounding [11].

The second contribution is a Gaussian process regression framework that exploits that mapping. The GP model starts from a conservative physics-based prior, learns the local FWHM-to-duration relationship from accumulated event data, produces explicit uncertainty estimates, and activates ride-through extension only when the predicted excursion duration — inflated by a conservatism factor — remains within the PRC-024 ride-through window. PMU-to-PMU observation sharing via the existing C37.118 layer expands the calibration dataset without placing communication latency in the real-time decision path. The local sovereignty architecture ensures that decisions are made by local power and the local state alone — a condition that must hold when UFLS thresholds are reached.

The Flagami case study — reconstructed by simulation — provides the evidentiary anchor: a well-documented cascade in which generating units tripped on frequency protection under conditions post-event analysis found consistent with successful ride-through. The simulation assesses whether the FWHM signature at those units is consistent with the duration predictions the architecture would have produced.

The translational objective is a provision pathway within NERC PRC-024 for model-based alternative compliance. The paper does not claim the GP architecture is ready for deployment. It claims that the physical foundation is established, that the duration-to-duration formulation is more direct and more tractable than an inertia-mediated one, that the V&V framework is adequate for protecting the credibility of protection engineering, and that the standards pathway is the right vehicle for the broader validation the concept requires. Two deferred items carry the work forward: the PNNL simulation, which produces the quantitative Flagami results; and the NERC Standards Authorization Request, which opens the standards development process.

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Appendix: Reviewer Questions

The following questions were developed during the paper's analytical phase to identify areas where each contributor's expertise is most directly relevant. Co-authors should treat them as an invitation to reshape the sections they own; reviewers should treat them as a structured review framework.

Bob Cummings

Standards, Architecture, and Evidentiary Record

1. The paper targets PRC-024 as the revision vehicle. Are there precedents in existing NERC reliability standards for an alternative compliance pathway — one that allows a site-specific, model-based approach alongside the existing fixed threshold envelope — or would this require a more fundamental standards restructuring?

2. From your NERC disturbance analysis experience, how complete is the Flagami evidentiary record? Specifically, what generation trip data, frequency recordings, and post-event analysis documents are available through NERC's disturbance reporting system that could anchor the simulation reconstruction?

3. Beyond Flagami, are there NERC lessons-learned documents, alerts, or Level 2 or Level 3 disturbance reports from other cascading frequency events that should be incorporated as additional evidentiary anchors? The paper is stronger if Flagami illustrates a pattern rather than standing as a single case.

4. Realistically, what evidentiary threshold would NERC require before a PRC-024 revision petition on predictive ride-through extension would be taken seriously? A simulation plus a conceptual standards pathway, or something closer to a field demonstration?

5. Is a NERC standards development petition the right near-term vehicle, or should this go through a FERC rulemaking, a NERC white paper, or a NERC-FERC joint technical conference first? The paper needs to name the right pathway.

6. As co-author, which sections do you want to own — the standards pathway section, the disturbance analysis background, or both?

Jeff Dagle — Pacific Northwest National Laboratory

Grid Modeling, PMU Systems, and Simulation Capability

1. Would PNNL be willing to run the Flagami network simulation for this paper? If yes, what platform would be used — PSS/E, PowerWorld, PSCAD, or a PNNL proprietary tool — and what does the commitment look like in terms of timeline and resource?

2. Does PNNL have a suitable southeastern grid model from the 2008 timeframe, or would the Flagami reconstruction require building the Florida network from NERC reliability data? The answer affects how we frame the simulation's fidelity.

3. Is there any PMU data in the PNNL archive from the Florida or southeastern grid at the time of the Flagami event — even sparse coverage — that could be used to anchor or validate the simulation? The paper is committed to a simulation-based framing, but any empirical anchor strengthens the reconstruction.

4. Is the temporal resolution of the simulation platform's output fine enough to compute FWHM from the modeled frequency time-series? The FWHM calculation requires resolving the wavefront shape, which means sub-second resolution at minimum, preferably at PMU reporting rates of 30–60 frames per second.

5. Does PNNL have existing capability in Gaussian process regression applied to power systems data, or would the ML element need to be developed from scratch? If existing tooling is available, does it integrate with the simulation platform in a way that lets the paper demonstrate the calibration concept quantitatively?

6. As co-author, what is your preferred scope of contribution — the simulation section, the signal processing methodology, the ML element, or the full technical core?

Rob Hoffman

Protection Engineering — Technical Plausibility and Deployment Architecture

1. From a protection engineering standpoint, is the FWHM conjecture physically plausible? Specifically, does the wavefront shape as observed at a local PMU carry information about effective inertia in the propagation corridor in a way that is consistent with how frequency disturbances propagate through actual network topologies?

2. The paper proposes a Gaussian process model with a physics-based prior and bounded parameter updates as the V&V framework for the learning element. Is that sufficient to establish protection engineering credibility, or does continuous model updating—even with bounds—create a compliance problem under current protection system qualification standards?

3. What is the right deployment architecture? Does the ride-through extension concept require a firmware modification to the protection relay itself, a separate processing unit running in parallel and issuing a supervisory override, or a locally sovereign PMU-resident function as described in Section 4.6? The paper argues for the third category — do you agree that it is a defensible and distinct architectural classification?

4. In your experience, what is realistic C37.118 latency on actual utility WANs? The paper has resolved that PMU-PMU sharing operates in the calibration loop rather than the real-time decision loop — does that framing fully resolve the latency concern from a protection engineering standpoint, or are there edge cases where the distinction breaks down?

5. The conservative activation threshold — the model must estimate inertia above the trip threshold by a specified margin before extension activates, and in any ambiguous case, the existing PRC-024 setting governs — is this the right design principle for getting protection engineers comfortable with the concept, or does it need to be stated more formally in terms of a specific reliability metric?

6. Is there a protection engineering standards body or working group — beyond NERC PRC-024 — that should be in the room when this goes to the standards conversation? IEEE PSRC comes to mind, but you may know of a more directly relevant working group.

Jeff Roberts — Roberts & Folkers Associates, LLC

Relay Design, Aurora Precedent, and Locally Sovereign Protection Architecture

1. The paper describes the Aurora protection device as a close-prevention function: when the device detects an out-of-synchronism condition, it lifts the close coil lead, making the injurious reclose mechanically impossible without a trip command, operator action, or remote communication. The paper then draws the mirror-image analogy to the UFLS ride-through architecture: where Aurora prevents an injurious close, the UFLS device prevents an unnecessary open, both reading local physics and acting locally. Does this characterization accurately reflect the Aurora device's operating principle, and does the analogy to the ride-through architecture hold in your assessment?

2. Is the GP learning architecture — a Gaussian process regression model resident in the PMU, updated on a scheduled basis from local and neighbor event data — compatible with the computational and memory constraints of current-generation protective relay and PMU hardware? Or does the architecture require a dedicated co-processor alongside the relay, and if so, is that a meaningful barrier to deployment?

3. The paper proposes that the locally sovereign PMU-resident function is a distinct architectural category from relay firmware modification and supervisory override. Is that distinction defensible to a protection engineering audience in the context of a NERC standards proceeding, and how should the category be defined in standards language to make the distinction enforceable?

4. Are there SEL relay or PMU families in current field deployment whose hardware could execute the FWHM computation and GP posterior inference within existing processing constraints at the required speed — a decision in the first hundreds of milliseconds of a disturbance — or does the architecture require hardware that does not yet exist in the field?

5. The V&V framework in Section 4.5 proposes bounded parameter updates, holdout validation, and audit logging as the qualification framework for the learning element. From your relay design experience, are these requirements achievable within existing protection system qualification standards — specifically IEEE C37.90 and applicable IEC standards — or do they require a new qualification category that would itself need a separate standards pathway?

6. As a reviewer, which sections are you best positioned to comment on, and would you consider co-authorship? Your Aurora device co-development history and your SEL patent record on protective relay architecture are directly relevant to the paper's core claims, and your name on the paper would carry significant weight with the IEEE PSRC audience this paper is trying to reach.